

4. (a) A lithium **atom** is denoted by ${}^7_3\text{Li}$.

Complete the table for the lithium atom. One cell has already been completed for you.

[3]

Particles	Number in ${}^7_3\text{Li}$ atom
electrons	3
protons	
neutrons	
leptons	
baryons	
up quarks	
down quarks	

- (b) (i) State what a meson is, in terms of quarks.

[1]

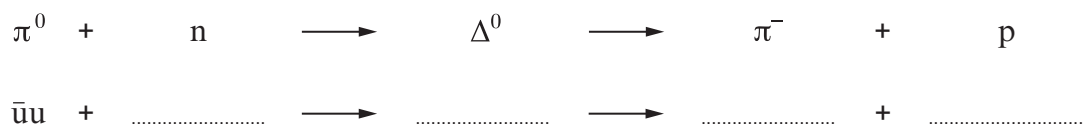
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- (ii) Explain why the quark make-up of a π^0 meson must be either $u\bar{u}$ or $d\bar{d}$.

[1]

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- (iii) The Δ^0 particle is a baryon which has a charge equal to that of a neutron. The following series shows the production **and** decay reactions of the Δ^0 particle. The reactions are also partially shown as a flow of quarks.



Complete the reactions in terms of quarks. The π^0 has been done for you.

[4]



- (iv) Explain clearly how the up quark number is conserved in both reactions.
[Consider both reactions separately.] [2]

Production of Δ^0 :

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Decay of Δ^0 :

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- (c) Textbooks state that the mean lifetime of the Δ^0 particle is $(5.63 \pm 0.14) \times 10^{-24}$ s.
State **two** features of the decay which suggest that it is a **strong** interaction. [2]

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